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A Method for Determining Blood Flow and Oxygen Consumption in the Rat Brain

By

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Abstract

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Cerebral blood flow (CBF) and cerebral metabolic rate for oxygen (CMR_{O_2}) were measured in rats under nitrous oxide anaesthesia, using a ^{133}Xenon modification of the Kety and Schmidt inert gas technique with sampling of cerebral venous blood from the retroglenoid vein. Extracerebral contamination of the venous blood sampled was studied by comparing the rates at which the activity of ^{133}Xenon decreased in blood and tissues. Contamination was avoided by gentle compression of the contralateral retroglenoid vein during sampling. CBF and CMR_{O_2} of the rat brain were 80 ± 2 and 7.6 ± 0.2 ml·(100 g)$^{-1}$·min^{-1}, respectively. These values are about 25% lower than those previously obtained for cerebral cortical tissue under similar conditions. Induced hypercapnia ($Paco_2$ about 70 mm Hg) or hypocapnia ($Paco_2$ 15–20 mm Hg) gave rise to expected changes in CBF but did not alter CMR_{O_2}. The CMR_{O_2} of the rat brain is at least twice that of the human brain. This species difference, which is similar to that previously reported for the oxygen uptake of cerebral tissue *in vitro*, probably reflects on inverse relationship between brain weight and neuronal packing density.

Interpretation of data on cerebral metabolic rate for oxygen (CMR_{O_2}), as obtained from determinations of cerebral blood flow (CBF) and of arteriovenous differences in oxygen content (AVD_{O_2}), in various species is hampered by differences in methodology. Values in man have usually been obtained by the inert gas technique of Kety and Schmidt (1948) and, since cerebral venous blood is then obtained from the internal jugular vein, the values for CBF (about 50 ml·(100 g)$^{-1}$·min^{-1}) and CMR_{O_2} (about 3.3 ml·(100 g)$^{-1}$·min^{-1}) are representative for the brain as a whole (for reviews, see Kety 1950, 1957, Lassen 1959, Sokoloff 1959). A direct comparison between values obtained in man and in experimental animals is difficult to make since (1) measurements in animals require some form of anaesthesia, and (2) non-primates have venous outflow systems that do not readily allow sampling of venous blood from the brain without contamination by extracerebral blood. For this reason, most measurements in animals have been carried out with sampling of venous blood from the superior sagittal sinus. In the dog under light thiopental or nitrous oxide anes-

thesia, the CMR_{O_2} values obtained are about 5–6 $ml \cdot (100\ g)^{-1} \cdot min^{-1}$ (Homberger *et al.* 1946, Michenfelder and Theye 1968) and, in the rat, a value of 10–11 $ml \cdot (100\ g)^{-1} \cdot min^{-1}$ has recently been reported (Eklöf *et al.* 1973, Norberg and Siesjö 1974). Although these results indicate that appreciable species differences exist, the comparison is complicated by the fact that with sampling from the superior sagittal sinus the values probably reflect metabolic rate in predominantly cortical brain areas. Thus, it has remained unsettled to what degree the higher CMR_{O_2} values in experimental animals reflect a true species difference, or a higher metabolic rate in tissues drained by the superior sagittal sinus.

Clearly, a relevant comparison between man and experimental animals requires that the sites of venous sampling are comparable. Since measurements of "cortical" CMR_{O_2} cannot be performed in man it seems desirable to try to sample venous blood in experimental animals that represents the outflow of the brain as a whole. Anatomical studies demonstrate that many laboratory animals have a venous outflow system which is different from that of man or the monkey (Batson 1944, Hegedus and Shackelford 1965, Butler 1967). In dogs and rats the internal jugular vein is small. The major part of cerebral venous blood is drained via the two main branches of the transverse sinus, *viz.* the petrosquamous sinus and the sigmoid sinus. Blood in the sigmoid sinus is drained predominantly via the vertebral venous system, and not via the internal jugular vein. The petrosquamous sinus continues as the emissary vein which emerges at the retroglenoid foramen (Butler 1967). Thus, in the dog and the rat, blood samples at this site should represent mixed cerebral blood.

The values for "cortical" CBF and CMR_{O_2} in the rat, reported from this laboratory, were obtained with an inert gas method, using ^{133}Xenon and repeated sampling of arterial and cerebral venous blood from the superior sagittal sinus during desaturation (Eklöf *et al.* 1973, Norberg and Siesjö 1974). This methodology was recently applied to the rat brain, using the retroglenoid vein for sampling of cerebral venous blood (Nilsson 1974). The CBF and CMR_{O_2} values determined were about 75% of those previously obtained, using the superior sagittal sinus as the source of venous blood. The results thus show that, as expected, cortical tissues have a higher metabolic rate than the brain as a whole. However, since the CMR_{O_2} value obtained for rat brain was at least twice that reported for the human brain, species differences were apparent. The present communication gives a full account of the measurements of CBF and CMR_{O_2} in rat brain, and discusses the validity of such measurements, especially in relation to contamination of venous blood from extracerebral sources.

Methods

Preparation of animals

The experiments were performed on male Wistar rats (290–410 g) that had access to pellet food and tap water until operation. Anaesthesia was induced with 2–3% halothane to allow tracheotomy and immobilization with tubocurarine chloride (1 mg per kg i.v.). Halothane was then withdrawn and the animals were maintained artificially ventilated on 70% N_2O and 30% O_2, the $Paco_2$ being adjusted to about 35 mm Hg. In 8 animals the $Paco_2$ was either increased by addition of CO_2 to the gas mixture administered, or decreased by hyperventilation. Both femoral arteries were cannulated, one for blood pressure recording and the other for sampling of arterial blood. The sampling catheter was cut to a length of 2–3 cm to minimize errors due to "smearing". One femoral vein was cannulated for infusion of fresh donor blood during the period of blood sampling. The body temperature was kept at 37°C by means of intermittent heating.

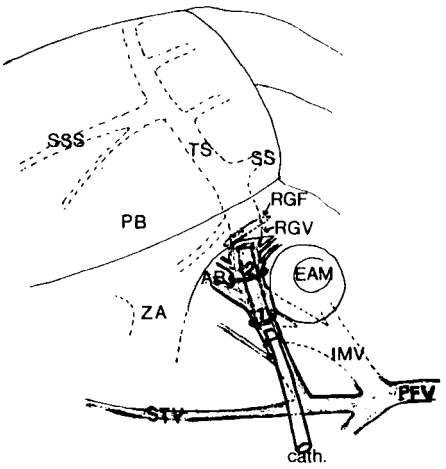


Fig. 1. Schematic representation of the dorsal cerebral venous outflow system in the rat as cannulated and ligated after partial resection of the temporal muscle. The possibilities of low and high ligation of the catheter are marked as 1 and 2, respectively (see text).

Abbreviations: SSS = Superior sagittal sinus; TS = Transverse sinus; SS = Sigmoid sinus; RGV = Retroglenoid vein; STV = Superficial temporal vein; IMV = Internal maxillary vein; PFV = Posterior facial vein; AB = Articular branch from temporo mandibular joint; RGF = Retroglenoid foramen; PB = Parietal bone; ZA = Zygomatic arch; EAM = External auditory meatus.

Preparation for venous sampling

The anatomy of the venous outflow system at the retroglenoid foramen was studied by means of retrograde silicone rubber injections into the external jugular vein (Fig. 1). The retroglenoid vein was cannulated via a superficial anastomosis. In the first part of the study the cannula was secured by a ligature (1) distal to a branch from the mandibular joint but, later, the ligature (2) was placed more closely to the exit of the retroglenoid vein through the bone. In order to study the effect of extracerebral contamination on the arteriovenous differences in ^{133}Xenon activity during desaturation, the external jugular vein was cannulated in 1 expt.

When the ligature was placed as indicated by (1) in Fig. 1 the ^{133}Xenon washout curves indicated the presence of extracerebral contamination. In order to minimize this contamination two changes were made. Firstly, the ligature was placed above the entrance of the joint branch (position 2, Fig. 1). Secondly, the contralateral retroglenoid vein was exposed and gently compressed during sampling of venous blood.

Measurements of CBF and $CMRO_2$

CBF and $CMRO_2$ were measured according to the principle described previously (see Norberg and Siesjö 1974). A bag with 5–10 mC of ^{133}Xenon in 30% O_2 and 70% N_2O was connected to the respirator, and saturation was continued for 20 or 30 min. At the end of the saturation period, samples were taken from artery and vein for ^{133}Xenon activity and total oxygen content (TO_2). The bag was then disconnected and repeated samples were taken from artery and vein for ^{133}Xenon activity. About 5 min after the beginning of desaturation a new set of arterial and venous samples were taken for TO_2 measurements. In order to avoid a fall in blood pressure due to repeated sampling, a slow infusion of fresh donor blood was given i.v.

All samples were collected in glass capillaries drawn to fine tips at both ends (capacity about 40 μl). The capillaries were connected to the catheter, and samples were obtained without suction. In those experiments in which compression of the contralateral sinus was performed, it was first controlled that blood flowed freely through the catheter. The contralateral sinus was then compressed, and when an increased outflow of blood was noted, the glass capillary was filled from the catheter. The rate of filling of the capillary was restricted by the lumina of the capillary tips and, if necessary, by gentle compression of the catheter so that the filling time was not less than 3 s. This was deemed necessary since it was found that if the flow through the catheter was too fast, extracerebral contamination occurred (see below).

The ^{133}Xenon activity was measured by gamma counting, and the CBF was calculated from the arterial and cerebral venous ^{133}Xenon desaturation curves, using the trapezoid rule. The tissue to blood partition coefficient used for ^{133}Xenon was 0.82 (see below). The TO_2 was measured by means of a polarographic technique (Borgström *et al.* 1974).

Tissue measurements

If extracerebral contamination of venous blood occurs, the venous blood will contain ^{133}Xenon even after cerebral structures have been desaturated (see Discussion). In order to allow estimation of desatura-

TABLE I. ^{133}Xenon partition coefficient (λ) calculated for different parts of the rat brain (activity per g tissue/activity per ml blood). Each figure represents one animal, and expresses the mean of 1–5 measurements on different pieces of tissue (number given in parenthesis).

Hemisphere			Mesencephalon	Cerebellum	Pons		Medulla
Cortex	Central-ventral parts	Overall			Overall	Ventral parts	
0.88 (3)	0.83 (2)	0.82 (5)	0.86 (2)	0.80 (1)	0.91 (1)	1.09 (1)	1.14 (1)
0.77 (2)	0.85 (2)	0.88 (5)	0.84 (1)	0.76 (1)		1.08 (1)	1.18 (1)
0.83 (5)		0.83 (3)					
0.82 (4)		0.79 (2)					
0.82 (4)		0.79 (2)					
0.75 (2)							
0.77 (5)							
0.80	0.84	0.83	0.85	0.78	0.91	1.09	1.16 (1)

tion of tissue, animals were decapitated after 10, 15 or 20 min of desaturation, following saturation periods of 20 or 30 min, and the heads were allowed to drop into liquid nitrogen. Various cerebral structures were grossly dissected with chisel and hammer under liquid nitrogen (cerebral cortex, central and basal parts of cerebral hemisphere, cerebellum, and brain stem), and placed in water-filled tubes for ^{133}Xenon counting (see Norberg and Siesjö 1974). The amount of tissue added to the tubes was determined by weighing.

Determination of the brain/blood partition coefficient for ^{133}Xenon

The partition coefficient was determined in 8 expts. Two animals were prepared as described above. In six animals the venous blood sampling was performed from the superior sagittal sinus. At the end of a 30 min saturation period arterial and cerebral venous blood samples were taken. The animal was then decapitated and the head immediately frozen in liquid nitrogen. The brain was chiselled out under liquid nitrogen and different parts were placed in water filled tubes for gamma counting.

Alternative estimation of mean brain flow

In two experiments mean brain flow was estimated from the tissue uptake of ^{14}C-ethanol, using a 30 s infusion period with repeated sampling of arterial blood and subsequent decapitation (Eklöf *et al.* 1974). The whole brain was then removed at $-10°C$, homogenized in distilled water, and aliquots of the homogenates were counted in liquid scintillation system together with the arterial blood samples. Mean CBF was calculated using the equation described by Reivich *et al.* (1969).

Results

The results of measurements of the partition coefficient for ^{133}Xenon, expressed as counts per g tissue/counts per ml cerebral venous blood (which is equal to arterial blood) are shown in Table I. The values for cortical tissue, central-basal noncortical tissue, and overall hemisphere tissue, averaged 0.80, 0.84, and 0.83, respectively. Since cortical tissue in the rat comprises about 70% of the hemisphere a partition coefficient of 0.82 was used for the calculation of CBF. It was assumed that some contribution of cerebellar and upper brain stem venous blood to the venous blood sampled should not significantly change this figure. The partition coefficients found are in good accordance with those given by Veall and Mallet (1965) for human cerebral cortex.

As stated, venous blood was first sampled without compression of the contralateral

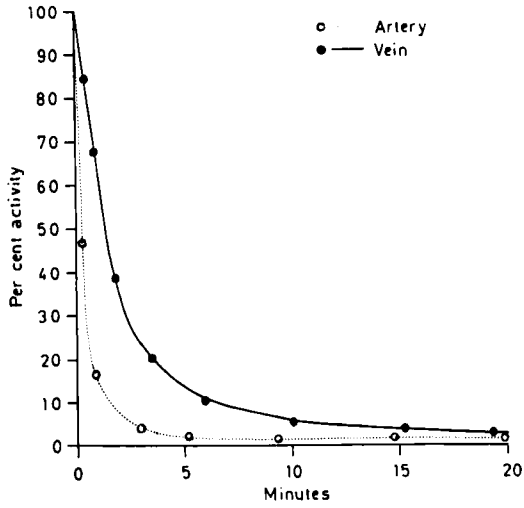


Fig. 2. Representative desaturation curves obtained without compression of contralateral retroglenoid vein and with ligature positioned as in (1) in Fig. 1. ^{133}Xenon activities expressed in percentage of venous activity at the end of saturation period. Note longlasting difference between arterial and venous activities.

transverse sinus. The desaturation curves obtained (Fig. 2) showed a distinct "tail", *i.e.* the venous ^{133}Xenon activity remained above that of the artery for long periods. The values for CBF and CMR_{O_2}, obtained in 5 animals, were 35–60 and 3.28–3.96 ml·(100 g)$^{-1}$·min^{-1}, respectively (ranges).

Theoretically, the slow decline of venous activity can be due either to the presence of slowly perfused brain tissue masses, or to contamination of the venous blood with blood from slowly perfused extracerebral regions (see Discussion). We tested for slowly perfused cerebral regions by saturating for 20–30 min, comparing the ^{133}Xenon activities in arterial

TABLE II. ^{133}Xenon activity in arterial and cerebral venous blood, compared to the activity in cerebral and extracerebral tissues. The animals were saturated during 20–30 min. The values express the per cent remaining activity after desaturation periods of 10, 15 or 20 min. In group 1, 2 and 3 the venous blood was sampled without compression of the contralateral retroglenoid vein (see text), while in group 4 the sampling was performed with compression, *i.e.* as in the final design of the CBF method.

Group	Desaturation time min	Activity (mean, range) in									
		Blood		Cerebral tissue					Extracerebral tissue		
		Arterial	Venous	Cortex	Central	Cerebellum	upper brainstem	lower brainstem	Muscle	Bone	Soft tissue
1	10	2.0	4.6	2.0	2.1	2.5	2.8	2.8	—	—	—
N=2		1.8–2.1	3.3–5.9	1.7–2.2	1.8–2.4	2.4–2.6	—	2.7–2.8			
2	15	1.0	2.9	0.9	1.2	1.2	—	2.0	—	—	—
N=2		0.8–1.2	2.7–3.0	0.8–0.9	1.0–1.4	1.2–1.2	—	1.8–2.2			
3	20	0.6	2.8	0.5	0.8	0.8	1.0	1.6	—	—	—
N=5		0.2–1.0	1.3–3.9	0.3–1.0	0.6–1.2	0.5–1.1	—	1.2–2.1			
4	15	0.8	1.0	0.7	1.0	0.9	0.8	1.7	17.0	10.5	16.4
N=5		0.4–1.2	0.8–1.2	0.4–1.2	0.9–1.3	0.6–1.3	0.2–1.3	1.6–1.8	13.6–20.6	7.1–13.8	13.0–19.8

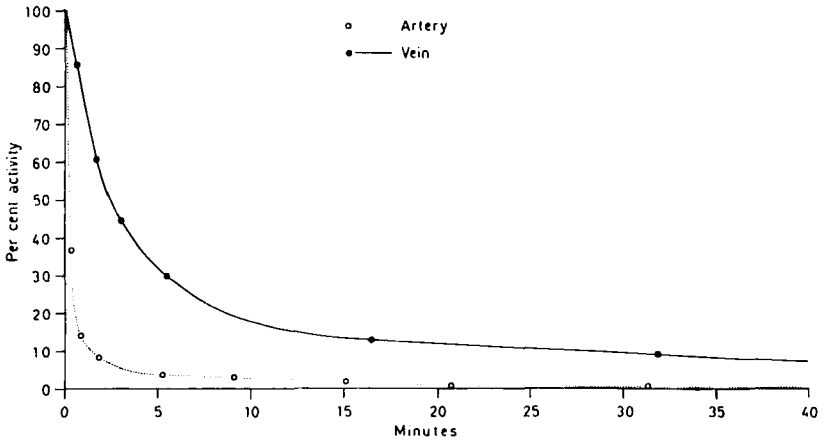


Fig. 3. ^{133}Xenon desaturation curves obtained with sampling of venous blood from external jugular vein. Period of saturation was 20 min. The prolonged arteriovenous difference reflects slow perfusion of extracerebral tissues (see text).

and cerebral venous blood to those in brain tissue, following desaturation for 10, 15 or 30 min. The results showed (Table II) that all brain regions except medulla oblongata reached an activity close to that of arterial blood after 15 min of desaturation, *i.e.* that slow perfusion of cerebral regions could not have contributed to the venous "tail".

Direct analyses of muscle tissue (masseter, neck muscles), bone (cranial vault, jaw) and skin (including hair), obtained after 15 min of desaturation, following a saturation period of 20 min, showed that such tissues still contained an activity 7–20% of that measured in arterial blood at the end of saturation. Thus, venous admixture with blood from such structures seemed responsible for the delay in desaturation of venous blood. By sampling venous blood from the external jugular vein (Fig. 3) an even more pronounced "tail" was observed, reflecting the slow perfusion of various structures in the head and neck.

In the second part of the experiments, compression of the contralateral sinus was performed to increase the flow of cerebral blood through the catheter, and the ligature around the catheter was placed close to the bone canal. Provided that the flow through the catheter was not too rapid (the capillary filled in less than 3 s) the desaturation curves regularly showed agreement between arterial and cerebral venous ^{133}Xenon activity within 10–15 min (Fig. 4). However, if a wide-bore capillary was attached to the catheter, flow through the catheter could amount to 2 ml per min, and exceed total flow through the brain (see below). In this case, the desaturation curves again showed delayed desaturation of venous blood, indicating contamination with extracerebral blood (Fig. 5).

Using compression of contralateral sinus, and controlled sampling of venous blood, 8 CBF and CMR_{O_2} determinations were performed at normal P_{CO_2} and another 8 at increased or decreased CO_2 tensions. The results (Table III) show that CBF was close to 80 and CMR_{O_2} about 7.5 $ml \cdot (100\ g)^{-1} \cdot min^{-1}$. In hypocapnia, CBF was reduced to 66% of normal and, in hypercapnia, CBF increased 3- to 6-fold, but the calculated CMR_{O_2} remained constant.

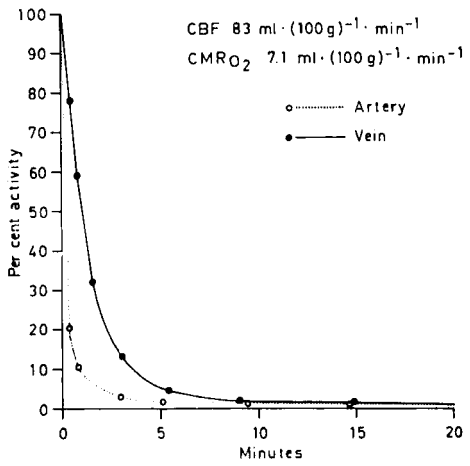


Fig. 4

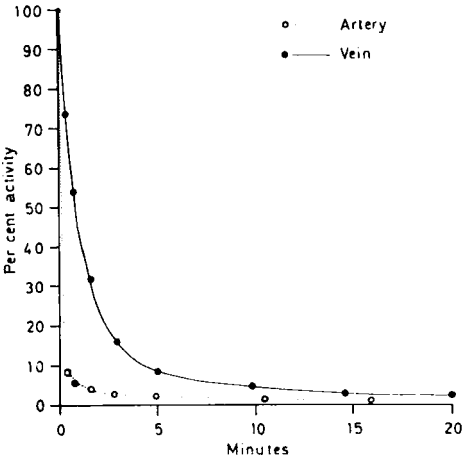


Fig. 5

Fig. 4. Representative desaturation curves obtained with compression of contralateral retroglenoid vein during venous sampling, and with catheter position as in (2) in Fig. 1. Note decrease in arteriovenous difference to zero within 10 min.

Fig. 5. ^{133}Xenon desaturation curves obtained with too rapid sampling of venous blood. The conditions were as in the example of Fig. 4 except for the fact that the use of wide-bore catheter and capillary permitted outflow at a rate of about 2 ml·min^{-1}.

In two experiments mean CBF, as calculated from the uptake of ^{14}C-ethanol in the whole hemisphere, was 79 and 80 ml·(100 g)$^{-1}$·min^{-1}. These results thus agree with those obtained using ^{133}Xenon and venous sampling.

Discussion

Before we discuss the physiological implications of the present results it seems warranted to scrutinize some problems that are inherent in the methodology of CBF and CMR_{O_2} measurements. CMR_{O_2} can only be meaningfully calculated if the CBF method measures CBF in an area whose venous outflow can be sampled. The inert gas technique of Kety and Schmidt (1948) is based on the Fick principle and quantitatively measures CBF provided

TABLE III. Mean arterial blood pressure (MABP), $Paco_2$, arterial oxygen content ($[To_2]a$), arteriovenous difference in oxygen content ($AVDo_2$), CBF and $CMRo_2$ in normocapnia, hypocapnia and hypercapnia. The values are means ±S.E.

Exp. group	MABP	$Paco_2$	$[To_2]a$	$AVDo_2$	CBF	$CMRo_2$
Normocapnia (n=8)	138 ±2	34.9 ±0.6	21.11 ±0.48	9.59 ±0.34	79.5 ±2.4	7.58 ±0.18
Hypocapnia (n=4)	146 ±7	16.8 ±1.6	22.61 ±0.79	14.44 ±1.08	52.2 ±4.7	7.42 ±0.40
Hypercapnia (n=4)	148 ±3	72 ±3	19.88 ±0.67	2.10 ±0.29	397 ±57	7.85 ±0.19

that certain requirements are fulfilled. The two most important of these are: (1) The tissue should not contain tissue masses with such low perfusion rates that saturation (or desaturation) times become unpractically long, and (2) the venous blood sampled should be of cerebral origin and there should be no significant contamination from extracerebral sources. It should be recognized that both these potential errors will give rise to a prolonged difference in inert gas activity between arterial and cerebral venous blood. If a "tail" is present, its cause is not readily apparent, and the error introduced is difficult to quantitate. The original Kety and Schmidt technique was criticized by Ferris *et al.* (1946) on grounds that extracerebral contamination was easily obtained, especially in hyperventilation, and by Sapirstein and Ogden (1956) on grounds that a persisting arteriovenous difference in nitrous oxide concentration due to slowly perfused tissue masses could introduce a considerable error in estimating flow. The work of Kety and associates (see Kety 1960) indicates that neither of these potential errors are serious when the methodology is applied to man. However, any new application of the method must take these potential errors into account.

When the ^{133}Xenon desaturation technique is applied to the rat, using the superior sagittal sinus for sampling venous blood, the venous activity approaches that of the artery already after 5–7 min (Eklöf *et al.* 1973, Norberg and Siesjö 1974). With sinus blood, neither slow perfusion of cerebral masses nor extracerebral contamination thus complicate the measurements. In the first part of the present series, each animal showed a delayed desaturation of venous blood, indicating that one or both of the potential errors discussed were involved. The results clearly showed that all cerebral regions with the possible exception of the lower brain stem had an activity close to that of arterial blood after 10 and 15 min of desaturation. In other words, slow perfusion of cerebral regions could not have been responsible for the venous tail, especially since the lower brain stem drains into spinal venous channels. This leaves contamination with extracerebral blood as the likely cause, and the presence of appreciable activities of ^{133}Xenon in muscle, bone and skin indicates that this was the case.

By increasing the flow of cerebral blood through the sinus catheter, compression of the contralateral sinus seemed to eliminate contamination with extracerebral blood. Several facts support this assumption: (1) In each animal the venous activity was close to that of the artery after 15 min of desaturation. (2) The CBF values obtained agreed with those derived from the tissue uptake of ^{14}C-ethanol, and (3) CMR_{O_2} remained constant in hypercapnia and hypocapnia although there were 6- to 8-fold differences in CBF.

Even if the contralateral sinus is compressed, precaution must be taken to prevent too large a flow. If a Pe 50 catheter and a wide-bore capillary are used, the outflow through the catheter may apparently exceed the total brain blood flow (mean flow = 80 ml·(100 g^{-1}·min^{-1}, brain weight = 2 g, total flow = 1.6 ml·min^{-1}). In such a case the low outflow resistance apparently allows diversion of extracerebral blood into the catheter. In practice, and by using flame-drawn capillaries, it is easy to control the catheter flow at such a rate that contamination does not occur.

It is a general assumption that the superior sagittal sinus drains mainly cortical areas, and that the CBF and CMR_{O_2} values, derived with the sagittal sinus as the source of venous blood, reflect grey matter flow and metabolism. The CBF values thus obtained are in excellent agreement with those calculated from the tissue uptake of ^{14}C-ethanol (Eklöf *et al.*

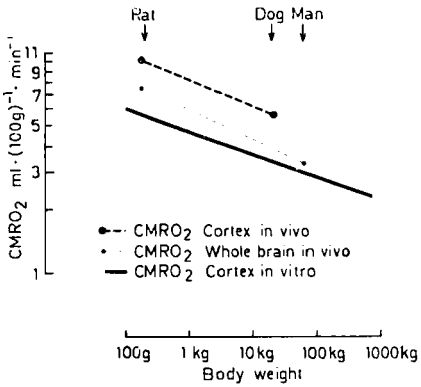


Fig. 6. Cerebral metabolic rate related to species body weight. The uninterrupted line illustrates the relationship between body weight and cerebral metabolic rate *in vitro*, as reported by Elliott and Henderson (1948). The thick stippled line gives the corresponding relationship for "cortical" CMR_{O_2} *in vivo*, obtained by joining the points for rat and dog, while the thin interrupted line shows the corresponding relationship for "whole brain" CMR_{O_2} in rat and man (see text).

1974, Salford and Siesjö 1974). In spite of this, the terms "cortical" CBF and CMR_{O_2} should be used with a clear understanding that also subcortical structures contribute to sagittal sinus blood (see Shenkin *et al.* 1948) and it should be recalled that cortical grey matter also shows inhomogeneities of flow (Freygang and Sokoloff 1958, Reivich *et al.* 1969). With these facts in mind we can conclude from the present experiments, and from our previous ones, that CBF and CMR_{O_2} in rats are higher in cortical tissue than in whole brain (by about 25 per cent).

The present results show that the CMR_{O_2} of the rat brain is at least twice as high as that of the human brain. Furthermore, cortical CMR_{O_2} in the rat seems to be twice as high as that in the dog (see Introduction). These differences probably reflect an inverse relationship between cerebral metabolic rate and body weight. This conclusion is in agreement with previous findings of differences in oxygen uptake by cortical tissue from various species *in vitro* (Elliott and Henderson 1948, Tower and Young 1973 a). This species difference in metabolic rate seems to reflect an inverse relationship between neuron density and brain weight. Although conflicting results have been published (Nurnberger and Gordon 1958), the majority of workers conclude from neuron counts, and from estimates of glia/neuron indices, that the relative number of neurons increases with decreasing brain size while there is a corresponding decrease in the glia/neuron ratio (Tower and Elliott 1952, Friede 1954, Hawkins and Olszewski 1957, Cragg 1967, Tower and Young 1973 b). This conclusion is also supported by experiments showing that components of the acetylcholine system show relative increases with decreasing brain size (Tower and Elliott 1952, Tower and Young 1973 b). Thus, if it is accepted that neurons have a higher respiratory rate than glial cells (Elliott and Heller 1957, Korey and Orchen 1959, Hess 1961) the relative decrease in neuron density in larger brains can readily explain the species differences in CMR_{O_2}.

At the present time, quantitative data for CMR_{O_2} in whole brain of lightly anaesthetized animals is only available for man and rat, and cortical CMR_{O_2} is only known for dog and rat. It has previously been found that if cortical oxygen uptake *in vitro* is plotted against brain weight on a log–log scale, a linear relationship is obtained (Elliott and Henderson 1948, Tower and Young 1973 a). If we assume that a similar relationship holds *in vivo* we can, by extrapolation, obtain tentative values for human cerebral cortex (Fig. 6). Thus,

CMR_{O_2} of human cerebral cortex should be about 4.5 $ml \cdot (100\,g)^{-1} \cdot min^{-1}$. This value is provisional and further quantitative comparisons must await quantitative measurements of CBF and CMR_{O_2} in different species.

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